Real World Advisory Memo for Options

Hedging a Concentrated Equity Position Using Put Options

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1 Client Situation

The client currently owns **500 shares** of a large-cap stock trading at **₹2,450 per share**, giving the position a market value of:

$$500 \times 2450 = ₹12{,}25{,}000$$

The client is concerned about a potential **10–15%** decline in the stock price over the next **three months**. However, the client does not wish to liquidate his position due to tax considerations and their long term positive outlook on the company. The objective is therefore to protect downside risk while retaining the ownership of shares.

2 Call vs. Put

A **put option** provides the holder the right, but not the obligation, to sell the underlying at a predetermined strike price (K).

$$\text{Put payoff at Maturity: } \max(K - S, 0)$$

2.1 Why a Put is Appropriate

Since the client fears a decline in stock price, a long put increases in value when the stock falls and therefore acts as insurance.

2.2 Why a Call is not Appropriate

A long call profits from bullish trends and does not provide protection against downside risk.

3 Strike Selection and Moneyness

I recommend purchasing a **3-month put** with strike **₹2400**. Current stock price :

$$S_0 = ₹2450$$

Strike:

$$K = ₹2400$$

This put is currently **Out-of-the-Money (OTM)** because:

$$K < S_0$$

An OTM put is cheaper than an ATM put and still offers ample protection against a **10–15%** correction.

3.1 ATM vs. OTM Discussion

- **ATM Put (₹2450 strike):**
  - Better protection.
  - Higher time value.
  - Higher Premium cost.
- **OTM Put (₹2400 strike):**
  - Lower premium.
  - Protection after a 2% decline.
  - More cost efficient.

3.2 Intrinsic vs. Time Value

Since the put is OTM:

- Intrinsic value ≈ 0.
- Most of the premium consists of **time value**, reflecting uncertainty and volatility during the next three months.

4 European vs. American Options

Indian exchange-traded stock and index options are generally traded as **European style** contracts, that is, exercise occurs only on expiration. Since Indian exchange-traded options are European-style, the client cannot exercise early and must plan the hedge duration carefully.

4.1 European Option

Can only be exercised on expiry.

4.2 American Option

Can be exercised at any time before expiry.

5 Payoff Table

Table 1: Hedged Portfolio Value at Expiry

Stock Price	Put Payoff/Share	Total Put Payoff	Stock Value	Portfolio Value
₹2100 (14% decline)	₹300	₹1,50,000	₹10,50,000	₹12,00,000
₹2300 (6% decline)	₹100	₹50,000	₹11,50,000	₹12,00,000
₹2450	₹0	₹0	₹12,25,000	₹12,25,000
₹2600	₹0	₹0	₹13,00,000	₹13,00,000
₹2700	₹0	₹0	₹13,50,000	₹13,50,000

6 Recommended Trade

Purchase:

- **3 month ₹2400 Put Option**
- Quantity sufficient to hedge 500 shares

Alternative:

- If the stock is highly correlated with the broader market, the client may purchase a **Nifty 50** protective put as a partial hedge, though stock-specific risk would remain.

The loss excluding the premiums does not exceed **₹25,000**. This is approximately 2% of the total portfolio value.

7 Key Risks

- **Theta Decay:** Option loses value as expiry approaches.
- **Wrong Strike Selection:** Excessively OTM strikes may provide inadequate protection.
- **Liquidity Risk:** Wide bid-ask spreads increase costs.
- **Opportunity Cost:** Premium paid reduces returns if the stock rises.

8 Learnings

One of the biggest things I learned from the payoff explorer was the difference between positions that initially seemed similar to me. Before working through the payoff diagrams, I was often confused between a long put and a short call, as both benefit when the underlying stock falls, and similarly between a long call and a short put, since both benefit when the stock rises. The visual payoff graphs helped me understand that while these positions may have similar directional views, their risk profiles are very different. A long put has limited downside (the premium paid) and substantial protection against falling prices, whereas a short call can have theoretically unlimited losses if the stock rises sharply. Likewise, a long call has limited risk and unlimited upside, while a short put exposes the seller to potentially large losses if the stock falls significantly. This distinction between market outlook and actual payoff structure was one of the most valuable insights I gained from the course and changed the way I think about option strategies.

9 References

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